

Data Visualization

Overview

Data visualization is an important step in exploratory data analysis, helping to identify trends, relationships, and patterns within the Human Development Index (HDI) dataset. In this project, the first 20 records of the dataset are used to generate clear and readable visualizations. These plots provide insights into how different socio-economic indicators influence HDI.

1. Unique Country Values

Objective

Identify all unique country names in the dataset to verify that each country appears only once and to ensure there are no duplicate records before performing further analysis.

Method

The `unique()` function is applied to the **Country** column, which returns every distinct country present in the dataset.

Observation

- Every country is listed only once.
- No duplicate country names were found in the selected dataset.
- This confirms that the data is suitable for visualization and analysis.

Result

The dataset contains unique country entries, ensuring accurate country-level comparisons throughout the project.

2. Mean Years of Schooling vs HDI

Objective

Analyze the relationship between the average years of schooling and the Human Development Index (HDI).

Visualization

A **Strip Plot** is used with:

- **X-axis:** Mean Years of Schooling
- **Y-axis:** HDI Score

Observation

- Countries with higher average years of schooling generally show higher HDI values.
- Most countries are concentrated within a narrow range, indicating consistent educational achievement among high-HDI nations.
- A few countries display slight variations, suggesting that education is an important but not the only factor affecting HDI.

Conclusion

Education has a positive association with human development, making it a valuable feature for HDI prediction.

3. Life Expectancy vs HDI

Objective

Examine how life expectancy influences the Human Development Index.

Visualization

A **Strip Plot** is created using:

- **X-axis:** Life Expectancy
- **Y-axis:** HDI Score

Observation

- Countries with longer life expectancy generally achieve higher HDI scores.
- The plot shows a positive trend, indicating improved healthcare and living conditions contribute to human development.
- Minor differences among countries suggest additional socio-economic factors also influence HDI.

Conclusion

Life expectancy is one of the significant indicators of human development and contributes positively to HDI.

4. Correlation Heatmap

Objective

Measure the relationship among numerical variables and identify the features most strongly associated with HDI.

Visualization

A correlation heatmap is generated using Seaborn, where each cell represents the correlation coefficient between two variables.

Observation

- Darker and lighter color intensities indicate the strength of negative and positive correlations.
- HDI shows strong positive relationships with indicators such as life expectancy, mean years of schooling, and gross national income.
- Highly correlated variables can serve as important input features for machine learning models.

Conclusion

The correlation heatmap helps identify the most influential variables, improving feature selection and model performance.

Overall Summary

The visualizations demonstrate that education, life expectancy, and economic indicators have meaningful relationships with the Human Development Index. These insights provide a strong foundation for feature selection, data understanding, and predictive model development in the Comprehensive Well-Being Framework.

